

Predictive Modeling of Cognitive Performance from Sleep Quality Metrics

CS 210: Data Management for Data Science · Anton Lyan · Rutgers University - New Brunswick · at1444@scarletmail.rutgers.edu

Main Goal(s):

The basic idea for this project was pretty straightforward: sleep affects how well you think, but most tools (i.e. apps, studies, even clinical assessments) just look at total hours slept. I wanted to see if going deeper into the actual structure of sleep (how much REM you get, how fragmented your sleep is, how long it takes you to fall asleep) could do a better job predicting cognitive performance than a single number.

Core question: do the individual components of sleep architecture each carry their own predictive signal, or is one composite score basically capturing everything?

Beyond the modeling question, the project also needed a proper data pipeline, not just a script that runs on a CSV, but actually storing the data in a relational database and querying it for analysis. Visuri et al. (2025) ran a two-month field study using wearable sleep rings and found that individual sleep metrics like latency, restfulness, and overall sleep quantity each independently predict cognitive performance. Their work supports the idea that collapsing sleep into a single score loses meaningful signal. This project takes a similar approach but applies it at scale using a regression pipeline rather than a controlled field study.

The Data

I used a sleep health dataset with 100,000 rows and 31 columns. Each row represents a person's sleep session along with their cognitive performance score (0–100) and a bunch of lifestyle and demographic variables.

The features break down into roughly four categories:

- Sleep architecture: duration, REM %, deep sleep %, sleep latency, wake episodes, nap duration
- Lifestyle: stress score, caffeine, alcohol, screen time, exercise, work hours, steps
- Physiological/demographic: age, BMI, resting heart rate, mental health condition, chronotype, shift work
- Contextual: occupation, country, season, day type, room temperature

One thing I had to be careful about was dropping two columns that looked like predictors but were actually concurrent outcomes: `sleep_disorder_risk` and `felt_rested`. Including either of those would've been data leakage, the model would've been learning from information that's essentially derived from the same moment as the target, not prior to it. I also dropped `person_id` since it's just an identifier.

The data was clean, no missing values, no duplicates. I did a $3\times\text{IQR}$ outlier check across all numeric features and found nothing extreme enough to remove. Since Ridge and Lasso are both robust to mild outliers anyway, retaining everything made sense.

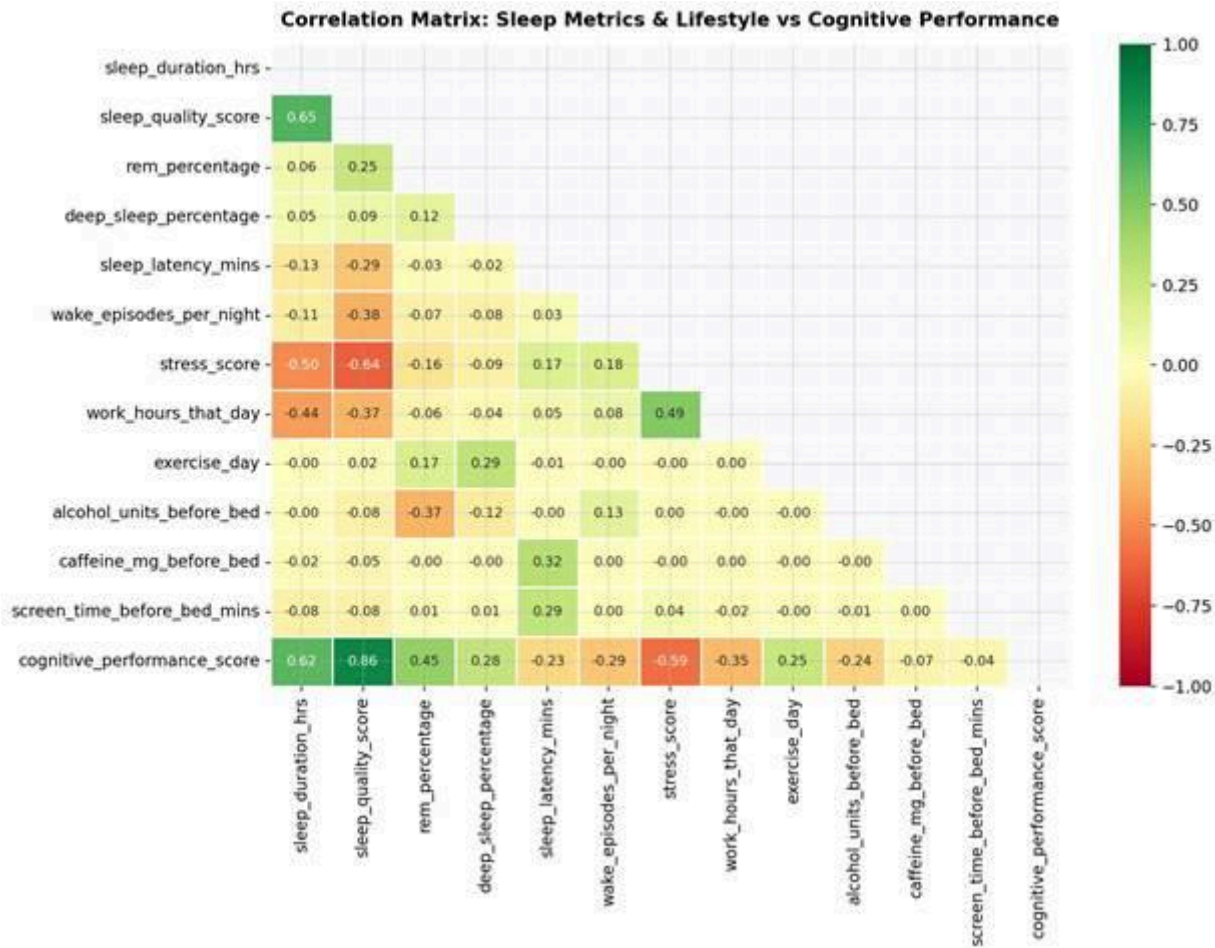


Figure 1. Correlation matrix of key sleep and lifestyle variables against cognitive performance score (bottom row). Stress score ($r = -0.59$) is the strongest negative predictor; sleep quality score ($r = +0.86$) is the strongest overall.

Database Setup

The raw CSV got loaded into a SQLite database with three normalized tables: participants, sleep_sessions, and lifestyle_factors. The idea was to separate the data by what it actually represents rather than keeping everything flat in one giant table.

Participants holds the demographic and baseline data. Sleep_sessions holds everything about the actual sleep event plus the cognitive score. Lifestyle_factors holds the behavioral context variables for that day. They're all linked by person_id.

All the downstream analysis: EDA, feature engineering, modeling, queries the database directly using a JOIN across all three tables. This felt a bit more work than just reading the CSV again, but it's the right pattern for a real pipeline and it meant the modeling code was decoupled from the file format entirely.

Feature Engineering

sleep_debt: how many hours short of 8 the person was. This is asymmetric: sleeping more than 8 hours doesn't get penalized, only sleeping less does. Captures the cumulative deficit framing from sleep science literature.

rem_ratio: REM as a proportion of structured sleep (REM + deep). The idea here is that the absolute REM percentage might be less informative than the balance between REM and deep sleep. Someone with 20% REM and 10% deep is in a different situation than someone with 20% REM and 25% deep.

sleep_fragmentation_index: wake episodes multiplied by sleep latency, divided by 100. This combines two disruption signals: how often you wake up and how hard it is to fall asleep in the first place. Both on their own are meaningful; together they capture a broader 'disrupted architecture' concept.

Modeling Approach

- I trained three models: plain linear regression as a baseline, then Ridge and Lasso. I ran each one on two feature sets: the full set (including the composite sleep_quality_score) and a sleep-metrics-only set (excluding that composite score). That split was intentional: it lets me directly measure how much predictive power comes from the raw architecture variables versus the aggregate.
- For train/test I did an 80/20 split by quartile of the target variable, so both splits have similar cognitive score distributions. All features were standardized before fitting, this is necessary for Ridge and Lasso because their regularization penalty is sensitive to feature scale. If you don't scale, features measured in large units (like steps_that_day) get over-penalized relative to features in small units (like alcohol_units_before_bed).
- For model selection I used RidgeCV and LassoCV, which do cross-validated lambda selection internally. On top of that I ran a separate 5-fold CV evaluation on the training set for each model to get stable estimates of MAE, RMSE, and R^2 with standard deviations across folds.

Results

Model	Feature Set	R^2	MAE	RMSE
Linear Regression	Full	~0.9303	~4.716	~5.918
Ridge Regression	Full	~0.9303	~4.716	~5.918
Lasso Regression	Full	~0.9303	~4.716	~5.918
Linear Regression	Sleep Only	~0.8885	~5.994	~7.484
Ridge Regression	Sleep Only	~0.8885	~5.994	~7.484
Lasso Regression	Sleep Only	~0.8885	~5.994	~7.484

The main noticeable comparison is that all three models perform identically on each feature set. That's not a bug, at 100k observations with relatively well-behaved features, OLS is already

stable enough that Ridge and Lasso have nothing to improve on. The regularization would matter more with a smaller dataset or stronger multicollinearity.

The more interesting number is the gap between feature sets. The full model hits $R^2 = 0.93$, but even without the composite sleep_quality_score, the raw metrics alone get to $R^2 = 0.89$. That 4% gap tells you that the composite score adds some signal, but the raw architecture variables are doing most of the heavy lifting. Individual sleep dimensions are genuinely informative, not just noise around a composite.

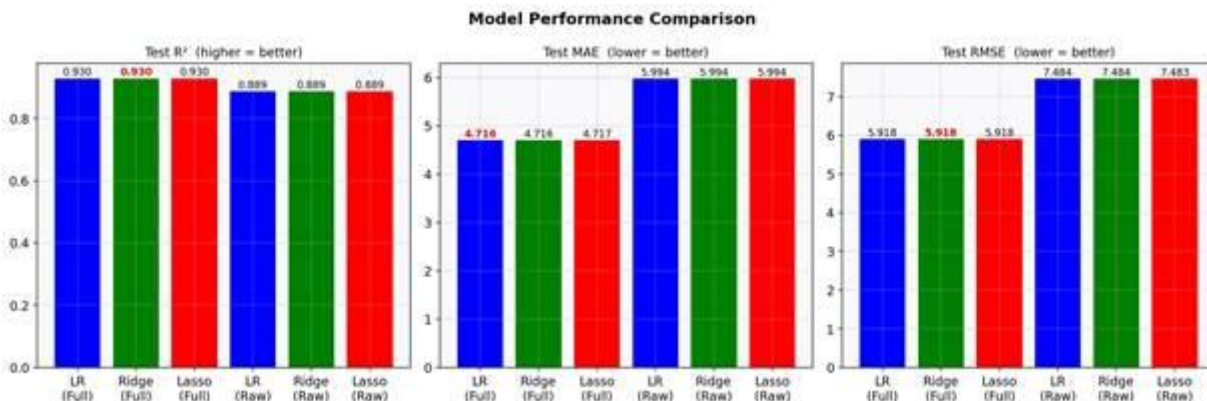


Figure 2. Model performance comparison. The gap between Full and Sleep-Only bars shows that raw sleep architecture alone ($R^2 = 0.89$) captures most of the predictive power even without the composite quality score.

In practical terms: the full model is off by an average of 4.7 points. The sleep-only model is off by 6.0 points. Both are pretty tight for a health prediction task. CV results closely matched test set results (difference < 0.001 on R^2), which confirms there's no overfitting.

What the Features Tell Us

Strongest positive predictors

Sleep quality score came out on top ($r = +0.86$), which isn't surprising given it's a composite. After that, sleep duration ($r = +0.62$) and REM percentage ($r = +0.45$) were the next strongest. Exercise and steps also showed up clearly as positive predictors, physical activity supporting cognition independent of sleep is a well-documented effect.

Strongest negative predictors

This is the more interesting side. Stress score ($r = -0.59$) was the single strongest negative predictor in the whole dataset, stronger in magnitude than any individual sleep architecture variable. That was genuinely surprising to me. It means cognitive performance is as much a stress problem as a sleep problem. If you're highly stressed, you're likely to score poorly on cognition even if your sleep metrics look decent.

After stress: work hours ($r = -0.35$), wake episodes ($r = -0.29$), shift work ($r = -0.25$), and alcohol before bed ($r = -0.24$). The alcohol finding makes sense, alcohol suppresses REM sleep, so its negative cognitive effect is partly mediated through sleep disruption.

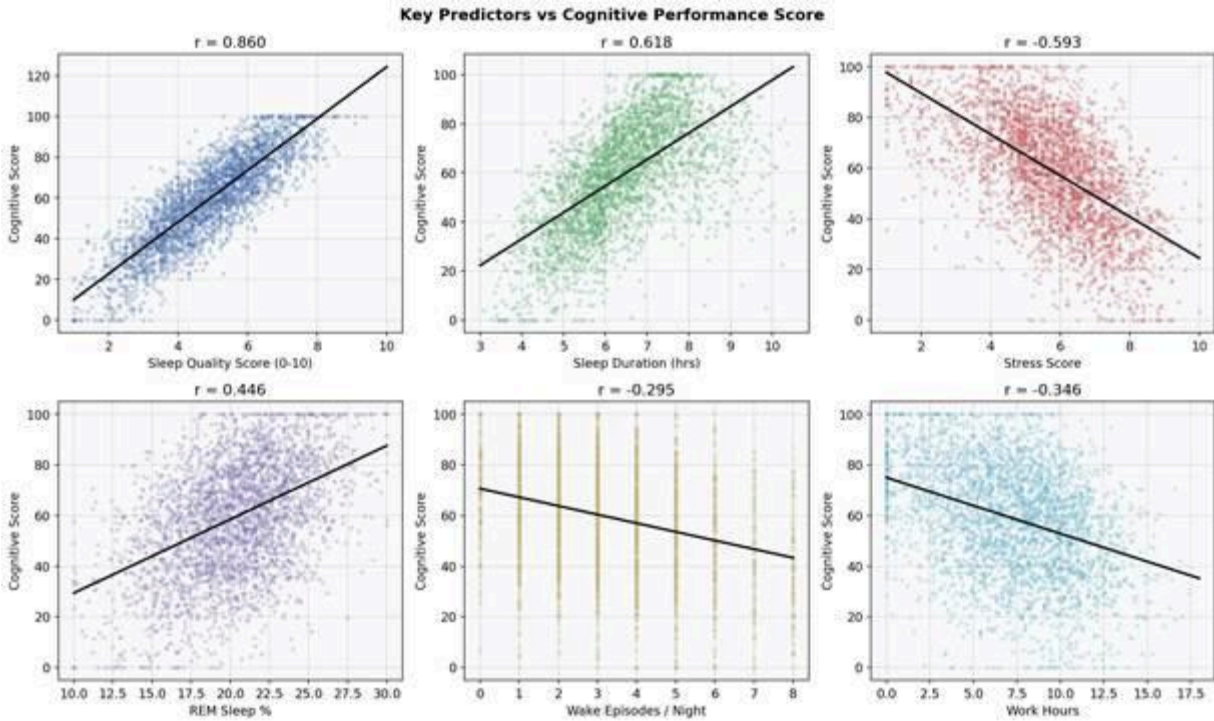


Figure 3. Key predictors vs cognitive performance score. The steep negative slope for stress ($r = -0.593$) is comparable in magnitude to the positive slope for sleep duration ($r = +0.618$), confirming that stress is as important a predictor as sleep itself.

Lasso feature selection

With the sleep-only feature set, Lasso didn't zero out any features at the selected lambda. That means every variable is contributing something once the others are in the model, there are no fully redundant predictors. It's a positive signal for the dataset: the features are capturing distinct dimensions rather than all measuring the same thing.

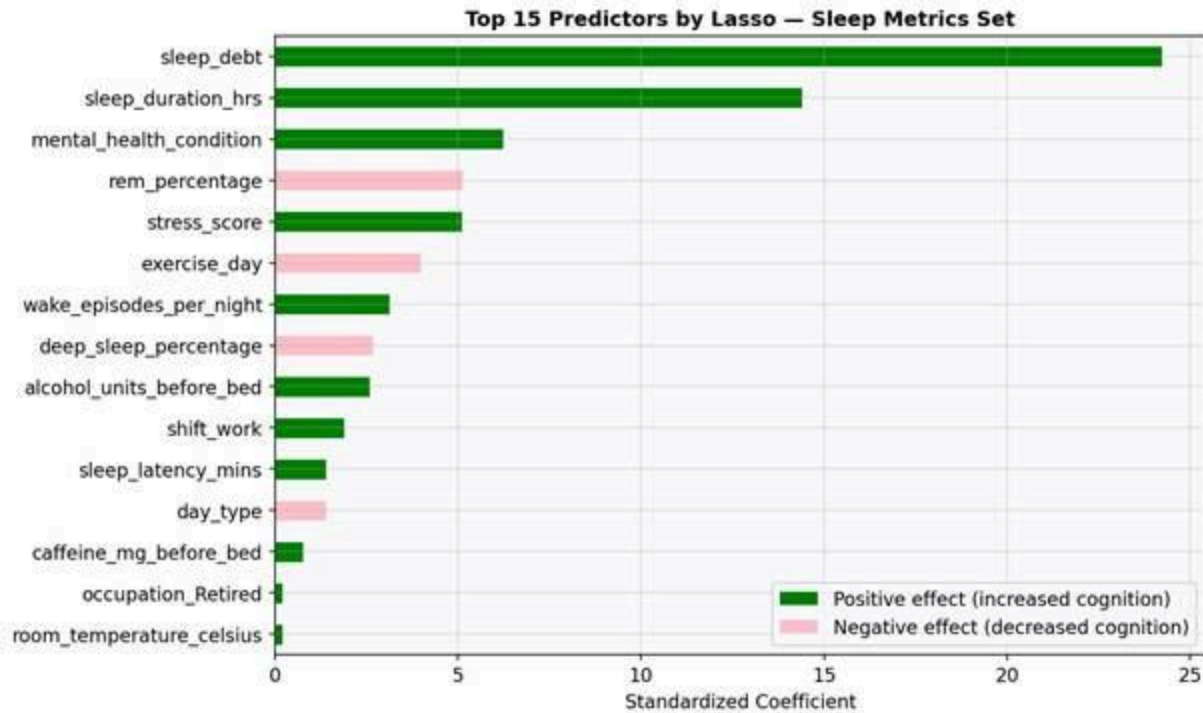


Figure 4. Top 15 predictors by Lasso coefficient magnitude (sleep metrics only). *sleep_debt* and *sleep_duration_hrs* dominate by a wide margin, confirming total sleep quantity is the strongest predictor in the raw feature set.

Residual Diagnostics

I ran four residual checks on the Ridge model: residuals vs fitted values, a Q-Q plot, a scale-location plot, and the residual distribution histogram.

The main findings: residuals are zero-mean (no bias), the distribution is approximately normal across the middle of the score range, and the Q-Q plot looks reasonable. The one thing worth noting is mild heteroscedasticity at the extremes, predictions near scores of 0 and 100 show slightly wider errors.

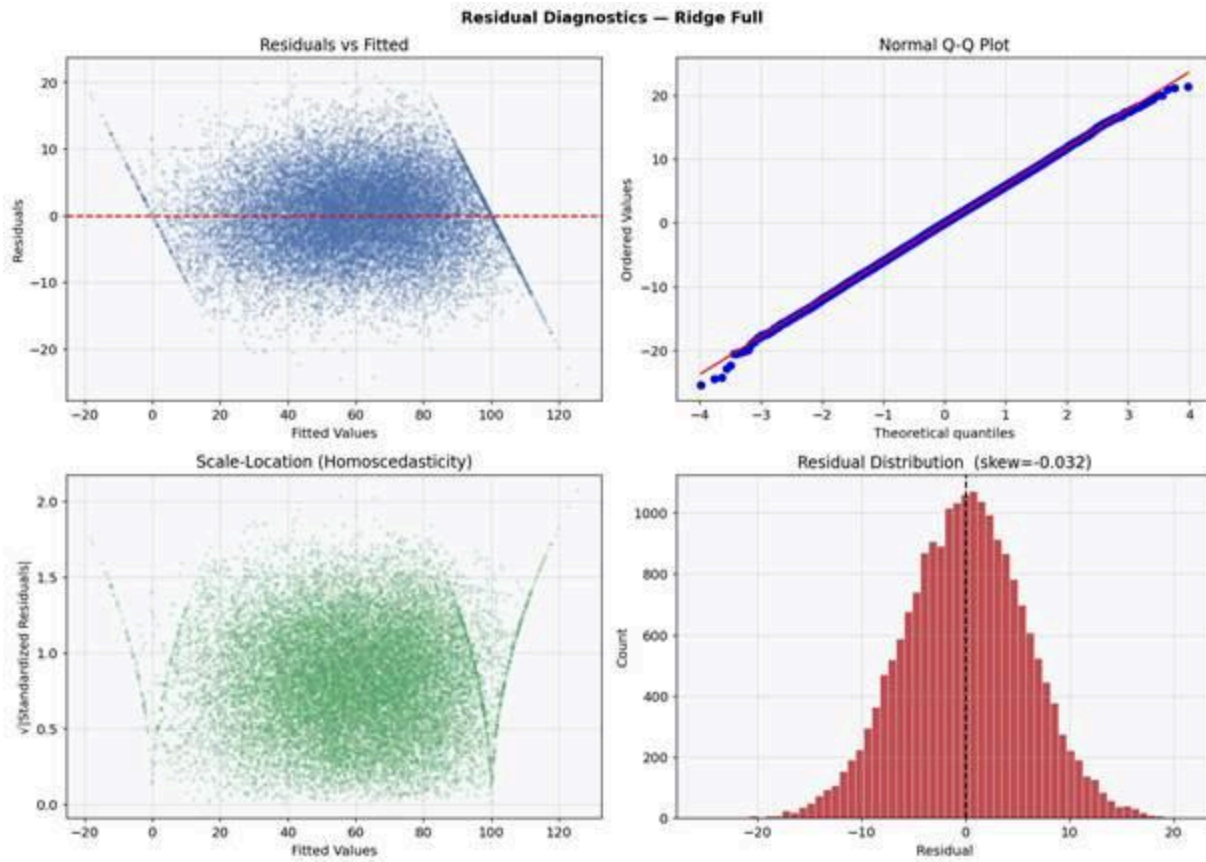


Figure 5. Residual diagnostics for the Ridge full-feature model. Residuals are centered at zero (no bias). The Q-Q plot confirms near-normality. The fan shape in the Scale-Location plot at fitted values near 0 and 100 shows the expected floor/ceiling heteroscedasticity. Residual skew = -0.032 .

Limitations

- The dataset is synthetic. The correlations are cleaner than you'd see in real wearable data collected from messy real-world conditions. An R^2 of 0.93 on real sleep data would be remarkable; on generated data it's expected.
- It's cross-sectional. Each row is a snapshot. You can't make causal claims from this, only associations. To actually establish that improving REM sleep causes better cognitive scores, you'd need longitudinal within-person data.
- No interaction terms. Stress $\times$ sleep_duration or alcohol $\times$ rem_percentage interactions could be meaningful but aren't modeled here.
- Residuals show Linear regression isn't the ideal tool for a bounded 0–100 outcome, logistic/beta regression could handle this better.

What I'd Do Differently

- Try a random forest model and compare against the linear results. My guess is it would pick up the nonlinear threshold effects (especially around sleep duration) and push R^2 higher.
- Model interaction terms more explicitly, particularly stress $\times$ sleep metrics, since the stress finding was the most interesting result and I'd want to understand whether high-stress individuals lose more cognitive function per unit of sleep deprivation.
- Test on a real dataset. Running this same pipeline on actual Fitbit or WHOOP exports would tell you how much the synthetic-data R^2 values hold up in practice.

Summary

The main takeaway is that raw sleep variables (not just a composite quality score) can explain ~89% of variance in cognitive performance scores, which validates the core hypothesis of the project. Modeling sleep as a multi-dimensional phenomenon is more informative than treating it as a single number.

The unexpected finding was the dominance of stress_score as a negative predictor. It outperformed every individual sleep metric, which suggests that any real-world cognitive performance prediction tool should probably weight psychological load as heavily as sleep architecture.

The SQL pipeline worked well for this task. Having the data in a normalized database rather than working directly from CSV made the feature engineering and modeling stages cleaner, and it's the right design pattern for anything that would eventually need to scale to real-time data.

References

Visuri, A., Koskimäki, H., van Berkel, N., Alorwu, A., Peltonen, E., Abdullah, S., & Hosio, S. (2025). *Cognitive performance measurements and the impact of sleep quality using wearable and mobile sensors*. arXiv preprint arXiv:2501.15583.